

| MODULE                                                                                                                           | LEARNING OBJECTIVES                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ACTIVITIES                                                                                                                                                                                                                                                                                                                                                   | ASSESSMENTS                                                                                                                                                                                                                                                                                                      |
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| Module 0<br><b>START HERE: Course Orientation</b>                                                                                | <ul style="list-style-type: none"> <li>● <b>0.1</b> Configure Canvas notification settings and verify access to JupyterLab, Python, and the course data files</li> <li>● <b>0.2</b> Examine course policies, grading criteria, and campus support resources to prepare for academic success</li> <li>● <b>0.3</b> Compose a reflective learning plan connecting personal goals and prior experience to course learning outcomes</li> </ul>                                           | <ul style="list-style-type: none"> <li>● Syllabus Review</li> <li>● Course policy review</li> <li>● Canvas navigation</li> <li>● JupyterLab and Python tools review</li> <li>● Academic Integrity</li> <li>● AI use overview</li> </ul>                                                                                                                      | <ul style="list-style-type: none"> <li>● <b>0.5 Question &amp; Discuss</b>  <b>Let's Get to Know Each Other</b> (intro. discussion)</li> <li>● <b>0.6 Confirm Your Course Readiness &amp; Syllabus Review</b></li> </ul>      |
| Module 1<br><b>Introduction to Data Analysis</b><br>Primary CLOs: <ul style="list-style-type: none"> <li>● I, II, VII</li> </ul> | <ul style="list-style-type: none"> <li>● <b>1.1</b> Explain the three main components of data science and how they work together in industry applications</li> <li>● <b>1.2</b> Launch JupyterLab and create a functional Python notebook for data analysis</li> <li>● <b>1.3</b> Use pandas to load a CSV dataset and perform basic DataFrame operations</li> </ul>                                                                                                                 | <ul style="list-style-type: none"> <li>● Read Sections: 1.1 - 1.7</li> <li>● Practice: Try It Yourself</li> <li>● Debug: Fix It </li> <li>● Answer: Quick Checks</li> <li>● Recall: Retrieval Practice</li> <li>● Review, Further Reading, and Looking Ahead</li> </ul>   | <ul style="list-style-type: none"> <li>● 1.6 Skills Lab 1A: <b>Your First Data Analysis Workflow</b> (Parts 1-3 and Questions &amp; Analysis)</li> </ul>                                                                                                                                                         |
| Module 2<br><b>Working with JupyterLab</b><br>Primary CLOs: <ul style="list-style-type: none"> <li>● VII</li> </ul>              | <ul style="list-style-type: none"> <li>● <b>2.1</b> Launch JupyterLab and create functional Python notebooks with code and Markdown cells</li> <li>● <b>2.2</b> Use keyboard shortcuts and Tab completion to work efficiently in JupyterLab</li> <li>● <b>2.3</b> Interpret error messages to identify and fix common Python coding errors</li> </ul>                                                                                                                                | <ul style="list-style-type: none"> <li>● Read Sections: 2.1 - 2.7</li> <li>● Practice: Try It Yourself</li> <li>● Debug: Fix It </li> <li>● Answer: Quick Checks</li> <li>● Recall: Retrieval Practice</li> <li>● Review, Further Reading, and Looking Ahead</li> </ul>   | <ul style="list-style-type: none"> <li>● 2.6 Skills Lab 2A: <b>Create a Professional Analysis Notebook</b> (Parts 1-3 and Q &amp; A)</li> </ul>                                                                                                                                                                  |
| Module 3<br><b>Data Structures and Control Flow</b><br>Primary CLOs: <ul style="list-style-type: none"> <li>● VII</li> </ul>     | <ul style="list-style-type: none"> <li>● <b>3.1</b> Create and manipulate lists, tuples, and dictionaries to store and organize data collections</li> <li>● <b>3.2</b> Implement conditional statements (if/elif/else) to control program flow based on different conditions</li> <li>● <b>3.3</b> Use for loops and while loops to iterate over data structures and repeat operations efficiently</li> </ul>                                                                        | <ul style="list-style-type: none"> <li>● Read Sections: 3.1 - 3.7</li> <li>● Practice: Try It Yourself</li> <li>● Debug: Fix It </li> <li>● Answer: Quick Checks</li> <li>● Recall: Retrieval Practice</li> <li>● Review, Further Reading, and Looking Ahead</li> </ul> | <ul style="list-style-type: none"> <li>● 3.6 Skills Lab 3A: <b>Building a Grade Analysis Tool</b> (Parts 1-3 and Questions &amp; Analysis)</li> <li>● <b>3.7 Question &amp; Discuss</b>  <b>Why Learn Loops?</b></li> </ul> |
| Module 4<br><b>Working with pandas DataFrames</b><br>Primary CLOs: <ul style="list-style-type: none"> <li>● II, IV</li> </ul>    | <ul style="list-style-type: none"> <li>● <b>4.1</b> Create pandas DataFrames from CSV files and Python data structures, understanding the relationship between dictionaries/lists and DataFrame structure</li> <li>● <b>4.2</b> Select specific columns and rows from DataFrames using bracket notation, .loc[], and .iloc[] indexing methods</li> <li>● <b>4.3</b> Filter DataFrame rows using .query() and boolean indexing to extract data subsets based on conditions</li> </ul> | <ul style="list-style-type: none"> <li>● Read Sections: 4.1 - 4.7</li> <li>● Practice: Try It Yourself</li> <li>● Debug: Fix It </li> <li>● Answer: Quick Checks</li> <li>● Recall: Retrieval Practice</li> <li>● Review, Further Reading, and Looking Ahead</li> </ul> | <ul style="list-style-type: none"> <li>● 4.6 Skills Lab 4A: <b>City Population Analysis</b> (Parts 1-3 and Questions &amp; Analysis)</li> </ul>                                                                                                                                                                  |

| MODULE                                                                                                                               | LEARNING OBJECTIVES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | ACTIVITIES                                                                                                                                                                                                                                                                                                                                       | ASSESSMENTS                                                                                                                                                                                                                                                                                                             |
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| Module 5<br><b>Data Exploration and Summary Statistics</b><br>Primary CLOs:<br><ul style="list-style-type: none"> <li>VII</li> </ul> | <ul style="list-style-type: none"> <li><b>5.1</b> Use <code>.info()</code>, <code>.describe()</code>, and <code>.value_counts()</code> to inspect DataFrame structure, generate statistical summaries, and identify data distributions</li> <li><b>5.2</b> Calculate and interpret measures of central tendency (mean, median, mode) and measures of spread (range, variance, standard deviation) to characterize dataset distributions</li> <li><b>5.3</b> Use <code>.groupby()</code> and aggregation methods to analyze data by categories and compare statistics across different subgroups</li> </ul>                                                    | <ul style="list-style-type: none"> <li>Read Sections: 5.1 - 5.7</li> <li>Practice: Try It Yourself</li> <li>Debug: Fix It </li> <li>Answer: Quick Checks</li> <li>Recall: Retrieval Practice</li> <li>Review, Further Reading, and Looking Ahead</li> </ul>   | <ul style="list-style-type: none"> <li><b>5.6 Skills Lab 5A: Analyzing Sales Performance Data</b> (Parts 1-3 and Questions &amp; Analysis)</li> <li><b>5.7 Question &amp; Discuss</b>  <b>The Warning Label</b></li> </ul>           |
| Module 6<br><b>Data Cleaning and Preparation</b><br>Primary CLOs:<br><ul style="list-style-type: none"> <li>II</li> </ul>            | <ul style="list-style-type: none"> <li><b>6.1</b> Identify missing data patterns using <code>.isnull()</code>, <code>.notnull()</code>, and <code>.info()</code></li> <li><b>6.2</b> Handle missing data appropriately using <code>.dropna()</code> and <code>.fillna()</code></li> <li><b>6.3</b> Convert data types, remove duplicates, and clean string data using pandas methods</li> </ul>                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>Read Sections: 6.1 - 6.7</li> <li>Practice: Try It Yourself</li> <li>Debug: Fix It </li> <li>Answer: Quick Checks</li> <li>Recall: Retrieval Practice</li> <li>Review, Further Reading, and Looking Ahead</li> </ul>   | <ul style="list-style-type: none"> <li><b>6.6 Skills Lab 6A: Cleaning Customer Survey Data</b> (Parts 1-3 and Questions &amp; Analysis)</li> </ul>                                                                                                                                                                      |
| Module 7<br><b>Data Visualization with matplotlib</b><br>Primary CLOs:<br>VI                                                         | <ul style="list-style-type: none"> <li><b>7.1</b> Create line plots, bar charts, and scatter plots using matplotlib.pyplot</li> <li><b>7.2</b> Customize plot appearance with titles, labels, colors, and styles</li> <li><b>7.3</b> Select appropriate visualization types based on data characteristics and analytical goals</li> </ul>                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Read Sections: 7.1 - 7.7</li> <li>Practice: Try It Yourself</li> <li>Debug: Fix It </li> <li>Answer: Quick Checks</li> <li>Recall: Retrieval Practice</li> <li>Review, Further Reading, and Looking Ahead</li> </ul>   | <ul style="list-style-type: none"> <li><b>7.6 Skills Lab 7A: Analyzing Temperature Trends</b> (Parts 1-3 and Questions &amp; Analysis)</li> <li><b>7.7 Question &amp; Discuss</b>  <b>The Chart That Lied</b></li> </ul>             |
| Module 8<br><b>Advanced Data Manipulation</b><br>Primary CLOs:<br>II                                                                 | <ul style="list-style-type: none"> <li><b>8.1</b> Sort DataFrames by single or multiple columns and create rankings to identify patterns in data</li> <li><b>8.2</b> Create pivot tables, cross-tabulations, and reshape data between wide and long formats to summarize and prepare multidimensional data</li> <li><b>8.3</b> Merge and concatenate multiple DataFrames to combine data from different sources</li> </ul>                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Read Sections: 8.1 - 8.7</li> <li>Practice: Try It Yourself</li> <li>Debug: Fix It </li> <li>Answer: Quick Checks</li> <li>Recall: Retrieval Practice</li> <li>Review, Further Reading, and Looking Ahead</li> </ul> | <ul style="list-style-type: none"> <li><b>8.6 Skills Lab 8A: Regional Sales Analysis Project</b> (Parts 1-3 and Questions &amp; Analysis)</li> </ul>                                                                                                                                                                    |
| Module 9<br><b>Introduction to Statistical Analysis</b><br>Primary CLOs:<br>VI, VII                                                  | <ul style="list-style-type: none"> <li><b>9.1</b> Calculate and interpret correlation coefficients using <code>.corr()</code> method and scatter plots to quantify the strength and direction of relationships between numeric variables</li> <li><b>9.2</b> Explain the logic of hypothesis testing including null hypotheses, alternative hypotheses, and p-values, and determine statistical significance at common alpha levels</li> <li><b>9.3</b> Compare groups using independent samples t-tests and chi-square tests (<code>scipy.stats.chi2_contingency()</code>), interpreting the resulting p-values to judge statistical significance</li> </ul> | <ul style="list-style-type: none"> <li>Read Sections: 9.1 - 9.7</li> <li>Practice: Try It Yourself</li> <li>Debug: Fix It </li> <li>Answer: Quick Checks</li> <li>Recall: Retrieval Practice</li> <li>Review, Further Reading, and Looking Ahead</li> </ul> | <ul style="list-style-type: none"> <li><b>9.6 Skills Lab 9A: Statistical Analysis of Education Data</b> (Parts 1-3 and Questions &amp; Analysis)</li> <li><b>9.7 Question &amp; Discuss</b>  <b>The P-Value Factory</b></li> </ul> |

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| Module 10<br><br><b>Functions and Code Reusability</b><br><br>Primary CLOs:<br><br>VII             | <ul style="list-style-type: none"> <li>● <b>10.1</b> Define and call functions with parameters and return values to perform reusable calculations</li> <li>● <b>10.2</b> Write docstrings following professional documentation standards to explain function purpose and usage</li> <li>● <b>10.3</b> Organize data analysis workflows into modular, maintainable functions</li> </ul>                                               | <ul style="list-style-type: none"> <li>● Read Sections: 10.1 - 10.7</li> <li>● Practice: Try It Yourself</li> <li>● Debug: Fix It </li> <li>● Answer: Quick Checks</li> <li>● Recall: Retrieval Practice</li> <li>● Review, Further Reading, and Looking Ahead</li> </ul>    | <ul style="list-style-type: none"> <li>● 10.6 Skills Lab 10A: <b>Building a Reusable Analysis Library</b> (Parts 1-3 and Questions &amp; Analysis)</li> </ul>                                                                                                                                                                                            |
| Module 11<br><br><b>Working with External Data Sources</b><br><br>Primary CLOs:<br><br>III, IV, V  | <ul style="list-style-type: none"> <li>● <b>11.1</b> Read and write data in multiple file formats including CSV, Excel, and JSON with appropriate parameters</li> <li>● <b>11.2</b> Use the requests library to fetch data from web APIs and convert responses into DataFrames</li> <li>● <b>11.3</b> Explain ethical considerations and best practices for web scraping and database access</li> </ul>                              | <ul style="list-style-type: none"> <li>● Read Sections: 11.1 - 11.7</li> <li>● Practice: Try It Yourself</li> <li>● Debug: Fix It </li> <li>● Answer: Quick Checks</li> <li>● Recall: Retrieval Practice</li> <li>● Review, Further Reading, and Looking Ahead</li> </ul>    | <ul style="list-style-type: none"> <li>● 11.6 Skills Lab 11A: <b>Multi-Source Data Integration Project</b> (Parts 1-3 and Questions &amp; Analysis)</li> <li>● <b>11.7 Question &amp; Discuss</b>  <b>The Last Data Analyst?</b> (Overall Course Concepts)</li> </ul> |
| Module 12<br><br><b>Ethics, Reproducibility, and Next Steps</b><br><br>Primary CLOs:<br><br>I, VII | <ul style="list-style-type: none"> <li>● <b>12.1</b> Identify potential sources of bias in datasets and analysis processes, and propose mitigation strategies</li> <li>● <b>12.2</b> Create reproducible analysis workflows using clear documentation, organized code, and version control</li> <li>● <b>12.3</b> Explain career pathways in data analytics and identify resources for continued professional development</li> </ul> | <ul style="list-style-type: none"> <li>● Read Sections: 12.1 - 12.7</li> <li>● Practice: Try It Yourself</li> <li>● Debug: Fix It </li> <li>● Answer: Quick Checks</li> <li>● Recall: Retrieval Practice</li> <li>● Review Questions</li> <li>● Course Conclusion</li> </ul> | <ul style="list-style-type: none"> <li>● <b>FINAL CAPSTONE:</b> <b>12.6 Skills Lab 12A Portfolio-Ready Analysis Project</b> (Parts 1-3 and Questions &amp; Analysis)</li> </ul>                                                                                                                                                                          |